

🔒 AES Encryption Test Cases

Select a test case to work through the AES encryption process step by step:

Select Test Case:

CBC | Key: 128 | "Hello, AES!"

Plaintext: Hello, AES!
Key (hex): 00112233445566778899aabbccddeeff
IV (hex): 0102030405060708090a0b0c0d0e0f10
Mode: CBC
Key Size: 128

Part I: Key Size and Mode

🔑 Key Configuration:

Determine the appropriate key size and encryption mode for the selected test case.

Key Size (bits):

128

Check your answer

Correct!

Mode:

CBC

Check your answer

Correct!

Part II: Key and IV/CTR

🔑 Encryption Parameters:

Enter the encryption key and initialization vector (IV) or counter (CTR) values based on the selected mode.

Key (hex):

00112233445566778899aabbccddeeff

Check your answer

Correct!

IV (hex):

0102030405060708090a0b0c0d0e0f10

Check your answer

Correct!

CTR (hex):

Enter CTR in hex

Check your answer

Not required for this mode.

Part III: Plaintext

 Input Text:

Enter the plaintext that will be encrypted using the AES algorithm.

Plaintext:

Hello, AES!

Check your answer

Correct!

Part IV: Plaintext Hex

 Hexadecimal Conversion:

Convert the plaintext to hexadecimal format with appropriate padding for block cipher requirements.

Plaintext (hex, padded):

48656c6c6f2c20414553210505050505

Check your answer

Correct!

Part V: Ciphertext

 Final Encryption Result:

Enter the final encrypted ciphertext in hexadecimal format after applying the AES algorithm.

Ciphertext (hex):

8ea2b7ca516745bfeafc49904b496089

Check Answer

Check your answer

Correct!